Dear Quillstone Payments Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. I bring six years of experience in analytics and data science, including more than four years developing and operationalising Python machine-learning models and scheduled GCP data workflows used by business teams. Quillstone's focus on helping merchants get quick, low-friction support particularly interests me, and I would welcome the opportunity to contribute my applied modelling and production-data experience to that work.

At Thistlebox, I developed a weekly LightGBM demand-forecasting model for orders by recipe and region, forecasting up to six weeks ahead. Feature engineering for calendar, price and menu-position factors helped reduce WAPE from 23% to 14%, with an estimated 9% reduction in food waste. I implemented a scheduled workflow in Vertex AI Workbench that writes forecasts to BigQuery for operational use in Looker. I also built an XGBoost churn model scored weekly in BigQuery to target retention offers; a 10% holdout validation found churn decreased by 2.1 percentage points in the treated group.

Alongside modelling, I design and analyse around 40 A/B tests each year and built a Python and Streamlit sequential-testing calculator with Docker, now used by around 25 colleagues. The data platform team handles its deployment on Cloud Run. I also write production SQL in BigQuery and maintain dbt models. These experiences have taught me to connect model development with scheduled data workflows, operational reporting, experimentation decisions and cross-team adoption, while working across teams and communicating results to leadership.

I hold an MSc in Statistics with distinction and bring a strong foundation in Python, statistical modelling and feature engineering, as well as hands-on experience with Google Cloud Platform data workflows, including scheduled Vertex AI Workbench jobs, BigQuery scoring and production SQL, and dbt analytics models. I would be glad to discuss how this background could support Quillstone's work on AI systems for customer operations and merchant support. Thank you for your consideration.

Yours sincerely,
Tobias Mwangi-Hartley